Pre-A Q3LOCK Twentieth-Moment, Edge-Cluster, and M2 Response-Contract Checkpoint

TECT verification-first repository
claim `C6-SPACETIME-SIGNATURE`

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1 Purpose and scope

This single gate-level checkpoint synthesizes `EXP-000811` and `EXP-000812`. It advances reusable result `R-167 v2.1` and reusable result `R-168 v1.2` without changing a claim card or tier. Both results are `T0`, additive, and `claim_bearing: false`. Their stable result identifiers are

1. `PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT`;
2. `PA-ROUND1-PROSPECTIVE-HOLDOUT-FREEZE-PROTOCOL-AND-CURRENT-TREE-READINESS-AUDIT`.

R-167 v2.1 proves two narrow children. A uniform twentieth endpoint-coordinate moment over every declared translate, edge orientation, split mesh, partial history, source, and history orientation implies a growing fixed-edge corridor with an explicit polynomial cost. A separate conditional theorem reduces that moment to a local fifth Gibbs moment, a one-site quartic elliptic embedding, and simultaneous-bond fifth graph propagation. Independently, an exact two-site full-oscillator edge has a parity doublet separated from its third eigenvalue under explicit Schur or relative-form margins; nested parity-preserving onsite spectral Ritz restrictions remove the local cutoff. Neither child supplies the missing Q3 fifth-moment inputs, a many-edge oscillator interaction, or a thermodynamic gap.

R-168 v1.2 proves two narrow children. At finite regulated Lane-Q volume, a quadratic scalar source contact can change every second-order free-energy or ground-energy curvature while leaving the zero-source Hamiltonian, first source derivative, state, spectrum, and 48-component mathematical fingerprint unchanged. A separate fail-closed schema fixes the minimum syntax and local artifact-binding contract for a future physical-response successor. Its 57 hostile cases and 48 deterministic malformed/fuzz cases validate only that schema; no physical response, candidate, prediction, target, or freeze is created.

The exact historical v0.9 combined source/PDF is prior evidence only for R-167 v2.0 and R-168 v1.1. It is not current evidence for R-167 v2.1 or R-168 v1.2 and cannot satisfy this checkpoint. No per-lemma or intermediate PDF was issued. This v1.0 fragment is the one combined synthesis source for the present logical checkpoint; its sibling PDF may be issued only after all proof, formal-authority, component, integrated, source-form, freshness, dual-extraction, and render-review gates pass.

2 Closed children, negatives, and open gates

The two R-167 children closed by `EXP-000811` are exactly

1. `PA-CP1-ST8-Q3LOCK-TWO-ORIENTATION-TWENTIETH-MOMENT-FIXED-EDGE-CORRIDOR-REDUCTION`;

2. PA-CP1-ST8-Q3LOCK-FULL-OSCILLATOR-EDGE-BLOCK-PARITY-DOUBLET-CLUSTER-AND-UNIFORM-ONSITE-SPECTRAL-CUTOFF-REMOVAL.

Its two new scoped negative results are exactly

1. NG-2026-08-11-PRE-A-ST8-Q3LOCK-UNIFORM-QUADRATIC-IN-M-ALL-MOMENT-BOND-SHEAR-GRAPH-TRANSPORT;
2. NG-2026-08-11-PRE-A-ST8-Q3LOCK-STATIC-MOMENTS-AND-LOW-GRAPH-AUTOMATIC-TWENTIETH-HISTORY-MOMENT.

The two R-168 children closed by EXP-000812 are exactly

1. PA-M2-CI8-LINEAR-PROBE-SECOND-ORDER-RESPONSE-NONIDENTIFIABILITY;
2. PA-M2-CI8-PHYSICAL-RESPONSE-SUCCESSOR-MINIMUM-CONTRACT-SCHEMA.

Its new scoped negative is exactly

NG-2026-08-11-PRE-A-M2-LANE-Q-LINEAR-SOURCE-AUTOMATIC-PHYSICAL-STIFFNESS-RESPONSE.

The two new mathematical input gates remain open:

1. PA-CP1-ST8-Q3LOCK-TRANSLATE-UNIFORM-LOCAL-FIFTH-GIBBS-MOMENT-AND-ELLIPTIC-EMBEDDING;
2. PA-CP1-ST8-Q3LOCK-SIMULTANEOUS-BOND-SHEAR-FIFTH-GRAPH-PROPAGATION.

The mathematical, physical-response, and prospective parents remain open:

1. PA-CP1-ST8-Q3LOCK-LOCAL-STRICT-ALL-EXHAUSTION-TWO-ORIENTATION-HISTORY-COMMON-ALPHA;
2. PA-CP1-ST8-Q3LOCK-INFINITE-DIMENSIONAL-RANK-TWO-BAND-BLOCK-DIAGONALIZATION-AND-TWO-PHASE-QPS;
3. PA-CP1-ST8-Q3LOCK-BROKEN-SECTOR-GNS-GAP-COERCIVITY;
4. PA-M2-CI8-PHYSICAL-RESPONSE-CHANNEL-AND-ERROR-BOUND;
5. PA-ROUND1-EVIDENCE-ROLE-AND-MINIMUM-MANIFEST-FREEZE;
6. PA-ROUND1-PER-PARAMETER-COMMON-INPUT-LEDGER;
7. PA-ROUND1-INDEPENDENT-CUSTODIAN-OPAQUE-TARGET-COMMITMENT;
8. PA-ROUND1-ADMISSIBLE-MICROSCOPIC-CANDIDATE-MAP-AND-FROZEN-PREDICTION;
9. PA-ROUND1-CRYPTOGRAPHIC-CUSTODIAN-SIGNATURE-AND-REMOTE-FREEZE-VERIFICATION.

3 Twentieth endpoint-coordinate moment corridor

For every tested translated edge $e = \langle xy \rangle$, finite volume, compact source, split mesh, partial history, and both history orientations, put

$$X_e = \max(|q_x|, |q_y|), \qquad M_{20} = \sup\{\sigma_+(X_e^{20}) + \sigma_-(X_e^{20})\}. \tag{3.1}$$

The supremum includes edge translates and orientations; translations alone still leave three cubic direction orbits. For the hard restricted edge tail

$$w_{e,L} = -c(q_x \cdot q_y)\mathbf{1}_{\{X_e > L\}}, \tag{3.2}$$

one has

$$w_{e,L}^2 \le c^2 X_e^4 \mathbf{1}_{\{X_e > L\}}. \tag{3.3}$$

After taking the summed two-orientation expectation,

$$\sigma_+(w_{e,L}^2) + \sigma_-(w_{e,L}^2) \le c^2 M_{20} L^{-16}. \tag{3.4}$$

For the induced edge set of $C_R = [-R, R]^3$,

$$m_R = 6R(2R+1)^2 \le 54R^3. \tag{3.5}$$

Quadratic-form Cauchy for $W_{R,L} = \sum_{e \in E_R} w_{e,L}$ therefore gives

$$\boxed{\sigma_+(W_{R,L}^2) + \sigma_-(W_{R,L}^2) \le 2916 c^2 M_{20} R^6 L^{-16}.} \tag{3.6}$$

More generally, a uniform $4p$-th moment and $L = R^\gamma$ give exponent

$$6 - 4\gamma(p-1). \tag{3.7}$$

The separate bounded-cutoff factorial requires $\gamma < 1/2$. Such a γ makes (3.6) negative if and only if $p > 4$, so the minimal integer is $p = 5$. The explicit choice $L = R^{2/5}$ yields

$$2916 c^2 M_{20} R^{-2/5} \longrightarrow 0, \qquad \log \frac{(CR^{4/5})^R}{R!} = -\frac{1}{5} R \log R + O(R). \tag{3.8}$$

This closes the conditional corridor implication only. It does not prove the actual Q3 onsite-interspersed twentieth history moment.

4 Conditional local-fifth-moment and fifth-graph theorem

Let $k_{x,h}$ be the positive shifted exact onsite Hamiltonian at source h. For a tested edge define

$$m_5 = \sup_{\Lambda, |h| \le h_0, x} \varphi_{\Lambda,h}(k_{x,h}^5), \qquad d_5 = \sup_{|h| \le h_0} \| \, |q|^{10} k_h^{-5/2} \|. \tag{4.1}$$

The second norm means the closed one-site quartic elliptic graph extension. For

$$\mu_z = e^{-\mu d(z,e)}, \quad K_e = \sum_z \mu_z k_{z,h}, \quad S_\mu = \sum_z \mu_z, \tag{4.2}$$

both endpoint weights are one and the distinct-site k_z strongly commute. Let $V_\times$ be the simultaneous all-cross-bond generator and $B_\delta = e^{-i\delta V_\times/\hbar}$. The load-bearing common-core or form-C^1 hypothesis is that the form domain of K_e^5 is invariant and

$$\frac{d}{d\delta}\langle B_\delta\psi, K_e^5 B_\delta\psi\rangle = \left\langle B_\delta\psi, \frac{i}{\hbar}[V_\times, K_e^5]B_\delta\psi\right\rangle. \tag{4.3}$$

With $J_e = (i/\hbar)[V_\times, K_e]$, require the uniform finite constant

$$G_5 = \left\| K_e^{-5/2}\frac{i}{\hbar}[V_\times, K_e^5]K_e^{-5/2}\right\|. \tag{4.4}$$

The five commutator positions pair with exact coefficients $1, 2, 2$:

$$\begin{aligned} G_5 \le\ & \|K_e^{-1/2}J_eK_e^{-1/2}\| \\ & + 2\|K_e^{1/2}J_eK_e^{-3/2}\| + 2\|K_e^{3/2}J_eK_e^{-5/2}\|. \end{aligned} \tag{4.5}$$

Gronwall gives

$$\|K_e^{5/2}B_\delta K_e^{-5/2}\| \le e^{G_5|\delta|/2}. \tag{4.6}$$

The onsite product flow commutes with K_e. A split prefix of total absolute bond time at most T, in either orientation, therefore has K_e^5 moment at most e^{G_5T} times the initial moment. Strong commutativity and scalar convexity give

$$\varphi(K_e^5) \le S_\mu^4\sum_z \mu_z\varphi(k_{z,h}^5) \le S_\mu^5 m_5, \tag{4.7}$$

while

$$X_e^{20} \le |q_x|^{20} + |q_y|^{20} \le d_5^2(k_x^5 + k_y^5) \le d_5^2K_e^5. \tag{4.8}$$

Hence the sharp two-orientation input is

$$\boxed{M_{20} \le 2d_5^2e^{G_5T}S_\mu^5m_5.} \tag{4.9}$$

The factor is 2, not 8: M_{20} already sums the two orientations, and the two endpoint fifth powers fit under one strongly commuting K_e^5. Neither the source/translate-uniform m_5, d_5 input nor the simultaneous-bond G_5 input has been proved for Q3; the two gates in Section 2 stay open.

5 Two dynamic no-go boundaries

The first negative rejects an automatic quadratic-in-m all-order graph hierarchy. Set $\hbar = 1$, $K = \mathrm{diag}(1, 4)$ and $V = \sigma_x$. Then

$$C_m = K^{-m/2}i[V, K^m]K^{-m/2}, \qquad \|C_m\| = 2^m - 2^{-m}. \tag{5.1}$$

For $A_m(t) = K^{m/2}e^{-itV}K^{-m/2}$, the weighted norm has a cusp and

$$\|A_m(t)\| = 1 + \frac{2^m - 2^{-m}}{2}|t| + O(t^2) \tag{5.2}$$

in the right-Dini, sign-maximized sense. Thus an $e^{G_m|t|/2}$ bound forces $G_m \ge 2^m - 2^{-m}$, divided by $\hbar$ in general. This does not reject the fixed $m = 5$ constant G_5.

The second negative rejects automatic promotion from static moments and only low graph powers. Let

$$K_N = \mathrm{diag}(1, N^4), \quad q_N = \mathrm{diag}(0, N), \quad H_N = \mathrm{diag}(0, N^4), \quad V_N = \frac{\sigma_y}{N^4}. \tag{5.3}$$

At $\beta = 1$, with $\rho_N \propto e^{-H_N}$ and $U_N(\delta) = e^{-i\delta V_N}$,

$$d_5 = 1, \qquad \rho_N(K_N^5) \le 1 + (5/e)^5, \tag{5.4}$$

and, for $0 \le s \le 1$ and $N^4 \ge |\delta|$,

$$\|K_N^s U_N(\delta) K_N^{-s}\| \le 1 + |\delta|. \tag{5.5}$$

Nevertheless

$$G_5 = N^6 - N^{-14}, \tag{5.6}$$

and each history orientation has q_N^{20} moment at least $\delta^2 N^{12}/8$; their sum is at least $\delta^2 N^{12}/4$. This is an automatic-inference counterfixture, not a Q3 dynamics nonexistence theorem.

6 Full-oscillator local-edge parity-doublet cluster

For coordination number $z = 6$, set

$$h_{xy} = \frac{(h_x - \epsilon_0) + (h_y - \epsilon_0)}{z} + B_{xy}. \tag{6.1}$$

Let P_0 be the two aligned low-product vectors, L the two misaligned low-product vectors, $P_{xy} = P_x P_y = P_0 + L$, and $Q = 1 - P_{xy}$. Define

$$\begin{gathered} C_b = 8c(a_0 - m^2), \qquad f_b = \frac{\delta_1}{z} + 4cd_2, \qquad e_b = C_b + f_b, \\ J = 8cm^2, \qquad \epsilon = 8c(b + 2ma + a^2), \qquad A = e_b + 2J, \qquad D = \frac{\Gamma}{z}. \end{gathered} \tag{6.2}$$

The exact statements are diagonal compressions,

$$P_0 h_{xy} P_0 = e_b P_0, \qquad L h_{xy} L = (e_b + 2J)L, \tag{6.3}$$

and

$$Q h_{xy} Q \ge DQ, \qquad \|Q h_{xy} P_{xy}\| \le \epsilon. \tag{6.4}$$

Neither P_0 nor L is asserted invariant. In general,

$$\|P_0 h_{xy} L\| = |f_b|. \tag{6.5}$$

This cross term need not vanish; it is nonzero in the rational fixture below. It does not enter the Rayleigh quotient on $P_0^\perp = L \oplus Q$, which is the only complement used below.

Assume that h_{xy} is nonnegative with compact resolvent, commutes with global parity, and satisfies

$$e_b \ge 0, \quad J > 0, \quad D > e_b, \quad \epsilon^2 < 2J(D - e_b). \tag{6.6}$$

The $L \oplus Q$ lower matrix has diagonal A, D and off-diagonal norm at most ϵ. Its lower eigenvalue is

$$g_b = \frac{A + D - \sqrt{(D - A)^2 + 4\epsilon^2}}{2} > e_b. \tag{6.7}$$

Opposite-parity aligned trials both have expectation e_b. Parity-sector min–max and (6.7) prove exactly one even and one odd eigenvalue at most e_b, together with

$$\lambda_3 \ge g_b, \qquad \lambda_3 - \lambda_2 \ge g_b - e_b. \tag{6.8}$$

When $D > A$,

$$g_b - e_b \geq 2J - \frac{\epsilon^2}{D - A}. \tag{6.9}$$

This is a local compression/min–max theorem. It is not a block diagonalization or a lattice QPS theorem.

7 Relative-form edge theorem and exact constants

Put

$$k = h_{\text{site}} - \epsilon_0 - \delta_1 P_1 \geq \Gamma Q, \qquad h_0 = \frac{k_x + k_y}{z} + J(1 - s_x s_y), \tag{7.1}$$

and

$$\gamma_0 = \min(\Gamma/z, 2J), \qquad \rho_b = \eta_b + \frac{\nu_b}{\Gamma}, \qquad \ell_b = 8c(a_0 - m^2) + 8c|d_2|. \tag{7.2}$$

For every $\tau > 0$, the registered diagonal-high and off-block bounds give

$$|\langle V \rangle| \leq \alpha \langle h_0 \rangle + \beta \|\psi\|^2, \tag{7.3}$$

where

$$\alpha = z \left[\rho_b + \frac{J}{\Gamma} + \frac{\epsilon^2}{\tau \Gamma} \right], \qquad \beta = \frac{2\delta_1}{z} + \ell_b + \tau. \tag{7.4}$$

If $\alpha < 1$ and

$$\Delta_{\text{rf}} = (1 - \alpha)\gamma_0 - 2\beta > 0, \tag{7.5}$$

then the same parity count holds and $\lambda_3 - \lambda_2 \geq \Delta_{\text{rf}}$.

The exact test fixture is

$$\begin{gathered} c = \frac{1}{1000}, \quad m = 2, \quad a = \frac{1}{10}, \quad b = \frac{1}{20}, \quad A_Q = 3, \quad a_0 - m^2 = \frac{1}{100}, \\ d_2 = -\frac{1}{1000}, \quad \delta_1 = \frac{1}{10000}, \quad \Gamma = 100, \quad z = 6. \end{gathered} \tag{7.6}$$

The negative sign of d_2 is load-bearing. Direct exact arithmetic gives

$$\begin{gathered} C_b = \frac{1}{12500}, \quad f_b = \frac{19}{1500000}, \quad e_b = \frac{139}{1500000}, \\ J = \frac{4}{125}, \quad \epsilon = \frac{23}{6250}, \quad A = \frac{96139}{1500000}, \quad D = \frac{50}{3}, \\ 2J(D - e_b) - \epsilon^2 = \frac{20832953}{19531250}, \\ g_b - e_b \geq \frac{332047248}{5188304375}. \end{gathered} \tag{7.7}$$

At the independent relative-form fixture $u = t = 1$ and $\tau = \epsilon$,

$$\eta_b = \frac{12}{125}, \quad \nu_b = \frac{402}{3125}, \quad \rho_b = \frac{15201}{156250}, \quad \alpha = \frac{183081}{312500}, \tag{7.8}$$

and

$$\beta = \frac{2851}{750000}, \quad \gamma_0 = \frac{8}{125}, \quad \Delta_{\mathrm{rf}} = \frac{4430237}{234375000} > 0. \tag{7.9}$$

The exact machine-oracle spellings are `20832953/19531250`, `332047248/5188304375`, and `4430237/234375000`. They are recomputed from the declared inputs, not pasted as derived constants.

8 Parity-preserving spectral cutoff removal

Let Π_M be nested parity-preserving onsite spectral projections with $\Pi_M \geq P$ whose union is a core for the quartic edge form. The valid finite restriction is the Ritz form restriction of the original h_{xy} to

$$(\Pi_M \otimes \Pi_M)\mathcal{H}_{xy}. \tag{8.1}$$

All constants in Sections 6–7 remain valid, and the Ritz eigenvalues decrease to the full edge eigenvalues. The lower gap is therefore uniform in M and passes to the full oscillator edge form.

This is not the Hamiltonian obtained by replacing q by $\Pi_M q \Pi_M$ before squaring. In general,

$$\Pi_M q^2 \Pi_M \neq (\Pi_M q \Pi_M)^2, \tag{8.2}$$

because the discarded virtual term is positive. A truncated-coordinate Hamiltonian need not preserve the exact compression or Ritz monotonicity.

In the registered large-N corridor,

$$e_b = O(N^{-6}), \quad \epsilon = O(N^{-3}), \quad D \asymp N^2, \quad 2J \longrightarrow 16, \tag{8.3}$$

with the mixing correction in (6.9) of order N^{-8}. Independently, $\alpha = O(N^{-2})$ and $\beta = O(N^{-3})$. These remain local two-site statements; no many-edge effective interaction or oscillator-lattice phase has been constructed.

9 Linear-probe second-order response nonidentifiability

Fix a finite-dimensional, finite-volume regulated Lane-Q Hamiltonian $H(t)$, a self-adjoint linear probe Q, a real source J, and a target-blind twice-differentiable scalar $d(t)$. Define

$$H_d(t,J) = H(t) - JQ + \frac{V}{2}d(t)J^2 I. \tag{9.1}$$

Every d gives

$$H_d(t,0) = H(t), \qquad \partial_J H_d(t,0) = -Q. \tag{9.2}$$

Thus the zero-source state, spectrum, field Hessian, and exact 48-component finite-torus fingerprint are unchanged. At finite β,

$$Z_d(t,J) = e^{-\beta V d(t) J^2/2} Z_0(t,J), \qquad F_{\beta,d} = F_{\beta,0} + \frac{V}{2}d(t)J^2. \tag{9.3}$$

With the positive helicity-like convention,

$$\Upsilon_d(t) = +\frac{1}{V}\partial_J^2 F_{\beta,d}(t,0), \qquad \Upsilon_d - \Upsilon_0 = +d(t). \tag{9.4}$$

The conventional scalar susceptibility $-V^{-1}\partial_J^2 F_\beta$ instead shifts by $-d(t)$. Wherever the ground branch is stable,

$$E_{0,d} = E_{0,0} + \frac{V}{2} d(t) J^2, \qquad \frac{1}{V} \partial_J^2 (E_{0,d} - E_{0,0}) = d(t). \tag{9.5}$$

The exact independent fixture uses

$$V = 7, \quad \beta = \frac{3}{2}, \quad h = \frac{1}{5}, \quad d_{\text{left}} = \frac{5}{7}, \quad d_{\text{right}} = \frac{11}{7}, \quad \Delta d = \frac{6}{7}, \tag{9.6}$$

and $H_0(J) = \text{diag}(-J, 4 + J)$. At $J = h$, the scalar-contact free-energy difference is 3/25 and the Boltzmann exponent shift is $-9/50$. The normalized centered curvature shift is 6/7; the ground indices at $J = -h, 0, h$ are $[0, 0, 0]$, and the two normalized ground-energy curvatures are 5/7 and 11/7.

This theorem proves nonidentifiability, not a contact law. A scalar $J\phi$ source is not automatically a physical helicity or superfluid-density probe. The compact or gauge action, quadratic contact, state/reference, control map, units, limit order, and error budget remain open.

10 Minimum physical-response successor contract and 57/48 audit

The closed schema child uses

`tect/pre-a-m2-ci8-physical-response-successor-minimum-contract/1.1`.

It is a schema-only result. The exact root field set is

```
schema, contract_id, candidate_id, parent_candidate_id,
status, fixture_only, candidate_created, version_delta,
physical_control_map, probe_contract, state_reference_contract,
response_definition, estimand_binding, critical_prediction,
error_budget, common_input_ledger, hard_row_rerun, verification,
prospective_firewall, no_overclaim
```

The declared nested schemas and types are exact. They cover the version delta, physical control map, full probe contract, state/reference contract, response definition, candidate-neutral estimand binding, critical prediction, six-term error budget, common-input ledger, ten-row rerun, three verifier roles, prospective firewall, and no-overclaim boundary.

Each artifact reference has exactly `path, sha256, role, media_type`. The path must be a normalized, case-exact, repository-relative POSIX path of at most 4096 characters, with no drive, backslash, dot segment, parent traversal, embedded NUL, trailing space, or trailing dot. The referenced file must exist, match its normalized SHA-256, and satisfy the role-specific root, suffix, and media-type rule. Path parsing, resolution, directory lookup, stat, hash, and open failures are structured rejections, not escaping exceptions.

A substantive successor declaration is a unique list drawn only from

`SECOND_ORDER_SOURCE_LAW`, `COMPACT_OR_GAUGE_ACTION`, `STATE_REFERENCE_CHANGE`, `PHYSICAL_CONTROL_MAP`, `REGULATOR_OR_LIMIT_CHANGE`, `ERROR_BOUND_PROOF`, and `MICROSCOPIC_MAP_ONLY`.

The first six entries are mandatory. A map-only singleton, missing mandatory entry, unknown or duplicate value, or a substantive label backed only by map-only content is rejected. The exact limit order is

`SOURCE_TO_ZERO, THERMODYNAMIC_LIMIT, REGULATOR_REMOVAL, CRITICAL_LIMIT_FROM_-ORDERED_SIDE`.

The physical control map and prediction fields reject target, holdout, discovery, and forbidden tokens. This finite lexical firewall is not a semantic target-independence proof.

The canonical positive-rational grammar is `[1-9][0-9]*(?:/[1-9][0-9]*)?`, with a maximum length of 128. Exact coprimality and canonical spelling are required; denominator-one slash forms such as `1/1` and `3/1` are rejected. Integer conversion failures, including a 5000-digit hostile literal, are caught in both engines.

Exactly six error terms are required: `finite_torus_spacing`, `regulator_removal`, `nonlinear_remainder`, `loop_or_renormalization`, `state_reference_transfer`, and `raw_estimator`. Every term binds an existing script, run JSON, and unique top-level result key, and the resulting `(script, run, result-key)` composites are distinct. At least two proof refs are distinct. The exact sum must be strictly below the common estimand margin. The ledger and section source IDs are unique and bound; the firewall allowed-source set is order-insensitive; and all ten rows `D00`–`D09` must be `PASS`. Primary, non-importing independent, and integrated refs are three distinct current Python files with three distinct actual hashes.

The positive syntax fixture uses six bounds $1/100$, so the total is $3/50 < 1/10$, and remains accepted after reversing the substantive changes, ledger rows, error terms, hard-row insertion order, and firewall sets. These ten PASS rows are syntax fixtures, not physical evidence.

The registered 57/48 schema audit has two count-pinned layers:

1. 57 exact hostile cases reject lifecycle promotion, map-only disguises, unbound or wrong-role artifacts, placeholder and target-bearing content, permuted limits, prediction mismatch, reused error evidence, missing result keys, verifier aliasing, ledger/source reuse, embedded-NUL and overlong paths, case/trailing-dot/trailing-space aliases, overlong or noncanonical rationals, and denominator-one slash forms with their required structured codes;

2. each engine rejects 48 deterministic malformed/fuzz cases, including embedded-NUL and overlong/alias paths, a 5000-digit rational, denominator-one rationals, malformed containers, non-string extra keys, and unhashable scalars, with all malformed cases rejected without exception.

Artifact existence and hashes do not prove semantic physical correctness, change-evidence equivalence, result-key fitness, script/run provenance equivalence, or source-class/used-for equivalence. The schema cannot promote the internal fingerprint into a physical prediction.

11 Current-candidate and prospective boundary

The 48 ordered finite-torus fingerprint components remain exactly one in the registered mathematical ordering, but this is not a physical dispersion prediction. The identifier `PA-M2-CI8-RS-DISPERSION-MAP-v1` remains `DESIGN_ONLY` and `NOT_CREATED`. No candidate, admitted microscopic map, physical response, controlled error enclosure, or frozen prediction exists.

The current source-only M2 extension still leaves D03, D05, D06, and D08 non-PASS; D07 still requires a real prospective holdout. A future freeze must complete the per-parameter common-input ledger, produce an independently committed opaque target, admit at least one target-blind microscopic candidate map with a nonempty prediction, and independently verify the custodian signature, remote commit, annotated tag, object, and ref. No target, custodian receipt, remote anchor, freeze, disclosure, score, or selection is created here.

12 Devil's-advocate

1. **UPHELD as an open input – twentieth moment:** the corridor theorem assumes uniform partial-history, translate, source, and orientation control. Static endpoint moments do not supply it.

2. **DISMISSED – factor eight:** the sharp conditional history bound has factor two because the definition already sums two orientations and both edge endpoints lie under one K_e^5.

3. **UPHELD as an open input – graph propagation:** the counterfixture has control through $0 \le s \le 1$, while the existing Q3 route reaches only its registered low rungs. Neither supplies the $5/2$ endpoint or the two upper commutator rungs.

4. **DISMISSED – all-order no-go:** exponential growth in the two-level all-m fixture rejects only an automatic polynomial hierarchy; it does not reject a fixed finite G_5.

5. **UPHELD – low-block invariance:** $P_0 h_{xy} L$ is generally nonzero. Only diagonal compressions and min–max on $P_0^\perp$ are used.

6. **DISMISSED – rational sign:** the fixture uses $d_2 = -1/1000$; the former positive sign was inconsistent with the declared a^2 bound. Every downstream fraction is recomputed with the negative sign.

7. **UPHELD – cutoff model:** a Ritz restriction of the full form is not the Hamiltonian obtained by truncating q before squaring.

8. **UPHELD – lattice transfer:** one full-oscillator edge cluster does not construct a quasi-local many-edge effective interaction, two-phase oscillator QPS system, or lattice GNS gap.

9. **DISMISSED – curvature sign:** the positive helicity-like curvature shifts by $+d$, while the conventional negative scalar susceptibility shifts by $-d$.

10. **UPHELD – physical meaning:** identical zero-source data and linear probe do not fix the quadratic contact or a physical stiffness response.

11. **DISMISSED – schema robustness:** both engines reject all 57 hostile cases and all 48 deterministic malformed/fuzz cases without an escaping exception, including embedded NUL, overlong paths, and a 5000-digit rational.

12. **UPHELD – schema semantics:** syntax, hashes, roles, and local result-key binding do not prove physical truth or create a candidate.

13. **UPHELD – parent scope:** four closed children do not close any common-alpha, many-edge rank-two, oscillator-gap, physical-response, prospective-freeze, C6, CP1, physical Sector-A, or Pre-A parent.

Independent adversarial review is invited especially on the unbounded form-C^1 hypothesis, the two extra fifth-commutator rungs, the $L \oplus Q$ lower matrix, Ritz form-core convergence, the source-contact sign convention, exact schema path confinement, rational parsing, evidence-role binding, and the semantic firewall.

13 Reproduction and evidence contract

The R-167 v2.1 primary, non-importing independent, and integrated scripts are

1. `codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split.py`;

2. `codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split_independent.py`;

3. codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split_verify.py.

The R-168 v1.2 primary, non-importing independent, and integrated scripts are

1. codes/foundations/pre_a_round1_prospective_holdout_freeze_protocol.py;
2. codes/foundations/pre_a_round1_prospective_holdout_freeze_protocol_independent.py;
3. codes/foundations/pre_a_round1_prospective_holdout_freeze_protocol_verify.py.

From the repository root, run

```
Set-Location E:\Dev\TECT
$py = 'E:\Dev\TECT.venv\Scripts\python.exe'
$r167 = (
  'codes\foundations\pre_a_cp1_st8_q3lock_' +
  'local_measured_renyi_semiclassical_doublet_route_split'
)
& $py -X utf8 ($r167 + '.py')
& $py -X utf8 ($r167 + '_independent.py')
& $py -X utf8 ($r167 + '_verify.py')
$r168 = (
  'codes\foundations\pre_a_round1_' +
  'prospective_holdout_freeze_protocol'
)
& $py -X utf8 ($r168 + '.py')
& $py -X utf8 ($r168 + '_independent.py')
& $py -X utf8 ($r168 + '_verify.py')
```

The final proof-layer outputs are R-167 primary `PASS 209/209`, independent `PASS 138/138`, and integrated `PASS 251/251`. The final R-168 outputs are primary `PASS 340/340`, independent `PASS 361/361`, and integrated `PASS 288/288`.

The six verification-script raw SHA-256 pins are

```
R-167 primary:
7162e22fe04eac53e7fe8adf3fb0d77a8f7374c70a5bcb3e5f96c06a53d7b868
R-167 independent:
6d46d3fea15dc17f1f783ac8a04d1f08f7c2c2ae2dea50fa809fee3f2b345104
R-167 integrated:
a62c23ca36ad7806800c83ed3821c92eb42a68fee3be1948aed6c0855385473f
R-168 primary:
78dd5b2bc79efe26e16f42f7c1f49ee93d550d8a9fdb65904f22d89729c7b7ec
R-168 independent:
313d5b76caf43dc0750a2d3cbbda245e1e492d370f9c5be4accd9dc66b25e8f4
R-168 integrated:
be735f01b7e42a9a71b5f92b73acfcfe9366679ec77300eb0ae6a69b39c18556
```

The integrated totals are final only after the same source/PDF metadata is pinned identically in both manifests, the PDF is fresh relative to this source, both pypdf and pdfplumber extract every declared page as nonempty and recover every required token, all rendered pages receive visual review, the stored component and integrated JSON files equal fresh invocations after normalization, formal authorities and generated surfaces are current, and the release check passes. A failed check invalidates the total rather than weakening the gate.

No PDF is built by this source-authoring step. No per-lemma or intermediate PDF belongs to R-167 v2.1 or R-168 v1.2. The eventual v1.0 PDF, if all gates pass, is the one current gate-level checkpoint; the historical v0.9 PDF remains strictly historical evidence.

14 Exact no-overclaim and dependency boundary

The common-alpha parent is **OPEN**. The twentieth-moment theorem is a conditional fixed-edge corridor reduction. It does not prove either new fifth-moment/graph input, the actual Q3 onsite-interspersed history moment, the $n \to \infty$ split limit, all-exhaustion Cauchy, the group, inverse or generator, or a fixed-beta KMS quotient.

The rank-two parent is **OPEN**. The full-oscillator result is one local edge compression/min–max and parity-preserving Ritz theorem. It does not give a linked-cluster or Lie–Schwinger many-edge elimination, a quasi-local effective interaction, or two-phase QPS for the oscillator lattice.

The spectral parent is **OPEN**. A local third-to-second edge gap is not an oscillator-lattice phasewise GNS gap, observed temporal mass, or physical mass gap.

The M2 physical-response parent is **OPEN**. Linear-probe nonidentifiability and a schema-only minimum contract provide no physical response channel, compact/gauge law, state/reference transfer, controlled error enclosure, or prediction.

The prospective parent is **OPEN**. No real prospective freeze, external target commitment, admitted candidate prediction, public remote anchor, authenticated receipt, disclosure, score, or selection exists.

Nothing in this checkpoint proves regulator removal, continuum control, same-Hamiltonian physical-empty comparison, physical vacuum selection, C6, CP1, physical Sector A, or Pre-A closure. The active `A5-SECTOR-A-SYNTHESIS` remains conditional at its registered scope.

15 Result footer

```
Result ID:              PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-
                        MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT;
                        PA-ROUND1-PROSPECTIVE-HOLDOUT-FREEZE-
                        PROTOCOL-AND-CURRENT-TREE-READINESS-AUDIT.
Reusable results:       R-167 v2.1; R-168 v1.2.
Explorations:           EXP-000811; EXP-000812.
Precise statement:      Conditional twentieth-moment fixed-edge corridor
                        with sharp fifth-moment/graph reduction; full-
                        oscillator local-edge parity doublet, gap, and Ritz
                        removal; fixed-linear-probe curvature
                        nonidentifiability; fail-closed minimum physical-
                        response successor syntax/binding schema.
Claim context:          C6-SPACETIME-SIGNATURE; claim_bearing false.
Closed children:        Two R-167 and two R-168 children, exactly the four
                        identifiers listed in Section 2.
Negative results:       Two R-167 and one R-168 scoped negatives, exactly
                        the three identifiers listed in Section 2.
Open input gates:       Local fifth Gibbs moment plus elliptic embedding;
                        simultaneous-bond fifth graph propagation.
Scope:                  T0 additive, claim-nonbearing analytic/exact result
                        at conditional history, local two-site oscillator,
                        finite regulated source-family, and schema-only
                        syntax/binding scope.
Dependencies:           R-167 v2.0; R-168 v1.1; EXP-000809; EXP-000810;
                        registered onsite doublet and finite-torus
                        fingerprint; open fifth-moment, fifth-graph,
                        many-edge, physical-response, custodian and remote
                        inputs.
Evidence grade:         ANALYTIC + EXACT + primary + non-importing
                        independent + integrated, conditional on the single
                        final checkpoint lifecycle.
Reproduction command:   Run the six commands in Section 13 from repository
                        root, primary then independent then integrated.
Expected output:        R-167: PASS 209/209,
```

	PASS 138/138, PASS 251/251. R-168: PASS 340/340, PASS 361/361, PASS 288/288.
Falsification gate:	Any moment exponent/factor, G5 rung, graph transport, Schur/min-max sign, parity count, rational constant, Ritz limit, contact-curvature sign, schema path/ rational/evidence binding, hostile/fuzz count, hash, assertion count, provenance, freshness, extraction, rendering, or scope mismatch.
Tier before / after:	C6 T1 / C6 T1; no claim or tier change.
No-overclaim statement:	No actual Q3 common alpha, many-edge rank-two oscillator elimination, oscillator-lattice GNS gap, physical response map or error bound, real prospective freeze, regulator removal, continuum, physical-empty sign, C6, CP1, physical Sector A, or Pre-A closure.
Next required action:	Prove both Q3 fifth-moment/graph inputs and the split limit; build a quasi-local many-edge oscillator transfer; derive a target-blind physical M2 response with a frozen error budget; obtain and verify a real prospective freeze before disclosure.
PDF policy:	No per-lemma or intermediate PDF. Historical v0.9 is prior evidence only; issue one v1.0 checkpoint only after every proof, formal, component, integrated, PDF, freshness, extraction, visual, and release gate passes.